

CW iter game

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Question1, finding indifference lines.

Consider the symmetric three-strategy game with payoff matrix

$$A = \begin{pmatrix} 2 & 0 & 1 \\ 1 & 2 & 0 \\ 0 & 1 & 2 \end{pmatrix}, \quad x = (x_1, x_2, x_3)^\top, \quad x_i \geq 0, \quad x_1 + x_2 + x_3 = 1.$$

The expected payoff to pure strategy i is

$$u_i(x) = (Ax)_i.$$

(a) Indifference lines

We determine the loci where two pure strategies give the same payoff.

Line $Z_{1,2}$:

$$(Ax)_1 = (Ax)_2 \iff 2x_1 + x_3 = x_1 + 2x_2 \iff x_1 = 2x_2 - x_3.$$

Using $x_1 + x_2 + x_3 = 1$,

$$1 - x_2 - x_3 = 2x_2 - x_3 \implies x_2 = \frac{1}{3}.$$

Line $Z_{2,3}$:

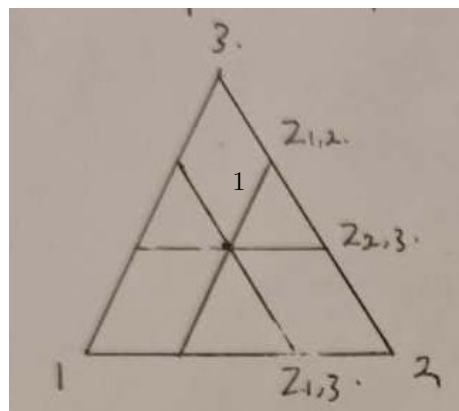
$$(Ax)_2 = (Ax)_3 \iff x_1 + 2x_2 = x_2 + 2x_3 \iff x_3 = \frac{1}{3}.$$

Line $Z_{1,3}$:

$$(Ax)_1 = (Ax)_3 \iff 2x_1 + x_3 = x_2 + 2x_3 \iff x_1 = \frac{1}{3}.$$

Thus the three indifference lines are

$$Z_{1,2} : x_2 = \frac{1}{3}, \quad Z_{2,3} : x_3 = \frac{1}{3}, \quad Z_{1,3} : x_1 = \frac{1}{3}.$$



(c) Best-response region for e_1

In the next step, we firstly find the best response region, such that we can find the Nash-Equilibrium points more clearly.

We firstly pick e_1 . The payoffs are

$$u_1(x) = 2x_1 + x_3, \quad u_2(x) = x_1 + 2x_2, \quad u_3(x) = x_2 + 2x_3.$$

To find the region in which e_1 is optimal, require

$$u_1(x) \geq u_2(x), \quad u_1(x) \geq u_3(x).$$

Condition $u_1(x) \geq u_2(x)$:

$$2x_1 + x_3 \geq x_1 + 2x_2 \iff x_1 + x_3 \geq 2x_2.$$

With $x_3 = 1 - x_1 - x_2$,

$$1 - x_2 \geq 2x_2 \iff x_2 \leq \frac{1}{3}.$$

Condition $u_1(x) \geq u_3(x)$:

$$2x_1 + x_3 \geq x_2 + 2x_3 \iff 2x_1 - x_2 - x_3 \geq 0.$$

With $x_3 = 1 - x_1 - x_2$,

$$3x_1 - 1 \geq 0 \iff x_1 \geq \frac{1}{3}.$$

Hence the best-response region for e_1 is

$$\{x \in \Delta : x_1 \geq \frac{1}{3}, x_2 \leq \frac{1}{3}\}.$$

By symmetry, we can derive the best-response regions for the other two pure strategies in the same way as for e_1 . Then we get: The best-response region for e_2 is

$$\{x \in \Delta : x_2 \geq \frac{1}{3}, x_3 \leq \frac{1}{3}\}.$$

the best-response region for e_3 is

$$\{x \in \Delta : x_1 \leq \frac{1}{3}, x_3 \geq \frac{1}{3}\}.$$

Combining the three regions for e_1, e_2, e_3 we obtain the usual partition of the simplex into three best-response cones, as illustrated in the phase diagram.

(e) Characterisation of all Nash equilibria

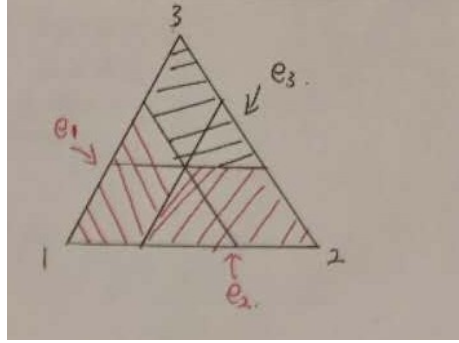


Figure 2: Best response region

From the best-response regions (Shown in figure 2) we can read off all symmetric Nash equilibria.

Pure-strategy equilibria. Each vertex e_i lies in the best-response region of the corresponding pure strategy:

$$e_i \in \text{BR}(e_i), \quad i = 1, 2, 3.$$

Hence all three pure strategies e_1, e_2, e_3 are Nash equilibria.

Two-strategy mixed equilibria on the edges. Consider, for example, the edge $\text{co}\{e_1, e_2\}$, where $x_3 = 0$. Any point on this edge can be written as

$$x = (p, 1 - p, 0), \quad 0 \leq p \leq 1.$$

For a symmetric Nash equilibrium supported on $\{1, 2\}$ we must have

$$u_1(x) = u_2(x) \geq u_3(x).$$

On this edge,

$$u_1(x) = 2p, \quad u_2(x) = p + 2(1 - p) = 2 - p, \quad u_3(x) = 1 - p.$$

The indifference condition $u_1(x) = u_2(x)$ gives

$$2p = 2 - p \implies p = \frac{2}{3}.$$

At this point,

$$u_1(x) = u_2(x) = \frac{4}{3}, \quad u_3(x) = \frac{1}{3},$$

so the third strategy is strictly worse. Therefore

$$\left(\frac{2}{3}, \frac{1}{3}, 0\right)$$

is a Nash equilibrium.

By cyclic symmetry of the game, we obtain two further edge equilibria

$$\left(0, \frac{2}{3}, \frac{1}{3}\right), \quad \left(\frac{1}{3}, 0, \frac{2}{3}\right),$$

located at the intersections of the indifference lines with the three edges of the simplex.

Interior equilibrium. In part (b) we found the unique interior point

$$\hat{x} = \left(\frac{1}{3}, \frac{1}{3}, \frac{1}{3}\right)$$

at which all three pure strategies are indifferent. Since every strategy is a best response to $\hat{x}$, this point is also a symmetric Nash equilibrium.

Collecting all cases, the set of (symmetric) Nash equilibria of this game is

$$\left\{e_1, e_2, e_3, \left(\frac{2}{3}, \frac{1}{3}, 0\right), \left(0, \frac{2}{3}, \frac{1}{3}\right), \left(\frac{1}{3}, 0, \frac{2}{3}\right), \left(\frac{1}{3}, \frac{1}{3}, \frac{1}{3}\right)\right\}.$$

2. Evolutionarily stable strategies (ESS)

We now determine which of the Nash equilibria found above are evolutionarily stable strategies (ESS). Recall the lemma we use later:

Lemma 1 (Lemma 1.3). *The ESS assumption (1.3) is equivalent to the condition that for all $y \neq x$ sufficiently close to x ,*

$$y^\top Ay < x^\top Ay. \quad (1.4)$$

Moreover, if $x \in \text{int } \Delta$ is an ESS, then

$$y^\top Ay < x^\top Ay \quad \text{for all } y \neq x.$$

Then, we will prove the following three circumstance

(a) The interior equilibrium is not ESS

Consider the interior Nash equilibrium

Firstly we have:

$$u(\hat{x}, x) = \hat{x}^\top Ax = 1$$

for every $x \in \Delta$

Now take the pure strategy $e_1 = (1, 0, 0)^\top$ as a contradiction, suppose $\hat{x}$ is an ESS. Its payoff against e_1 is

$$u(e_1, e_1) = e_1^\top Ae_1 = 2 \geq u(\hat{x}, e_1) = 1,$$

So by using lemma 1.3 we conclude it is not an ESS.

(b) The three edge equilibria are not ESS

Next consider the edge Nash equilibrium

$$x^{12} = \left(\frac{2}{3}, \frac{1}{3}, 0\right),$$

which lies on the edge $\text{co}\{e_1, e_2\}$. By symmetry, the same argument will apply to

$$x^{23} = \left(0, \frac{2}{3}, \frac{1}{3}\right), \quad x^{31} = \left(\frac{1}{3}, 0, \frac{2}{3}\right).$$

Take a nearby strategy of the form

$$y = \left(\frac{2}{3} + \varepsilon, \frac{1}{3} - \varepsilon, 0\right), \quad 0 < \varepsilon \ll 1,$$

which is still supported on $\{1, 2\}$ and close to x^{12} .

Then I want to show that $u(y, x^{12}) < u(y, y)$ to show that it is not an ESS.

We have $Ay =$

$$\begin{pmatrix} \frac{4}{3} + 2\varepsilon \\ \frac{4}{3} - \varepsilon \\ \frac{1}{3} - \varepsilon \end{pmatrix}.$$

Then

$$x^{12\top} Ay = \left(\frac{2}{3}, \frac{1}{3}, 0\right) \begin{pmatrix} \frac{4}{3} + 2\varepsilon \\ \frac{4}{3} - \varepsilon \\ \frac{1}{3} - \varepsilon \end{pmatrix} = \frac{4}{3} + \varepsilon,$$

while

$$\begin{aligned} y^\top Ay &= \left(\frac{2}{3} + \varepsilon, \frac{1}{3} - \varepsilon, 0\right) \begin{pmatrix} \frac{4}{3} + 2\varepsilon \\ \frac{4}{3} - \varepsilon \\ \frac{1}{3} - \varepsilon \end{pmatrix} \\ &= \left(\frac{2}{3} + \varepsilon\right) \left(\frac{4}{3} + 2\varepsilon\right) + \left(\frac{1}{3} - \varepsilon\right) \left(\frac{4}{3} - \varepsilon\right) \\ &= \frac{4}{3} + \varepsilon + 3\varepsilon^2. \end{aligned}$$

Such that:

$$y^\top Ay - x^{12\top} Ay = \left(\frac{4}{3} + \varepsilon + 3\varepsilon^2\right) - \left(\frac{4}{3} + \varepsilon\right) = 3\varepsilon^2 > 0$$

for every $\varepsilon > 0$. So we find a strategy y near $\hat{x}$ enough that not satisfy Lemma 1.3. So we conclude it is not an ESS.

By cyclic symmetry of the game, the same reasoning shows that

$$\left(0, \frac{2}{3}, \frac{1}{3}\right) \quad \text{and} \quad \left(\frac{1}{3}, 0, \frac{2}{3}\right)$$

are also Nash equilibria but not ESS.

(c) The pure strategies are ESS

We now verify that the three pure strategies $\{e_1\}, \{e_2\}, \{e_3\}$ are ESS. By symmetry it suffices to prove this for $e_1 = (1, 0, 0)$.

Consider a perturbed strategy of the form

$$y = (1 - \delta_1, \delta_2, \delta_1 - \delta_2), \quad \delta_1 \geq \delta_2 \geq 0,$$

with δ_1, δ_2 sufficiently small. Thus $y \in \Delta$ and $y \rightarrow e_1$ as $\delta_1, \delta_2 \rightarrow 0$.

By Lemma 1.3, if e_1 is ESS then for all such $y \neq e_1$ close to e_1 we must have

$$y^\top Ay < e_1^\top Ay.$$

First compute

$$Ay = \begin{pmatrix} 2 & 0 & 1 \\ 1 & 2 & 0 \\ 0 & 1 & 2 \end{pmatrix} \begin{pmatrix} 1 - \delta_1 \\ \delta_2 \\ \delta_1 - \delta_2 \end{pmatrix} = \begin{pmatrix} 2 - \delta_1 - \delta_2 \\ 1 - \delta_1 + 2\delta_2 \\ 2\delta_1 - \delta_2 \end{pmatrix}.$$

Hence

$$y^\top Ay = 3\delta_1^2 - 3\delta_1\delta_2 - 3\delta_1 + 3\delta_2^2 + 2,$$

while

$$e_1^\top Ay = 2 - \delta_1 - \delta_2.$$

Therefore

$$\begin{aligned} y^\top Ay - e_1^\top Ay &= (3\delta_1^2 - 3\delta_1\delta_2 - 3\delta_1 + 3\delta_2^2 + 2) - (2 - \delta_1 - \delta_2) \\ &= 3\delta_1^2 + 3\delta_2^2 - 3\delta_1\delta_2 - 2\delta_1 + \delta_2. \end{aligned}$$

Using $\delta_1 \geq \delta_2 \geq 0$, we have

$$-2\delta_1 + \delta_2 \leq -\delta_1,$$

and thus

$$y^\top Ay - e_1^\top Ay \leq 3\delta_1^2 + 3\delta_2^2 - 3\delta_1\delta_2 - \delta_1.$$

Moreover,

$$3\delta_1^2 + 3\delta_2^2 - 3\delta_1\delta_2 = O(\delta_1^2) + O(\delta_2^2) = o(\delta_1),$$

so there exists $\eta > 0$ such that whenever $0 < \delta_1 < \eta$ and $0 \leq \delta_2 \leq \delta_1$,

$$3\delta_1^2 + 3\delta_2^2 - 3\delta_1\delta_2 \leq \frac{1}{2}\delta_1.$$

Hence

$$y^\top Ay - e_1^\top Ay \leq \frac{1}{2}\delta_1 - \delta_1 = -\frac{1}{2}\delta_1 < 0,$$

showing that for every sufficiently small perturbation $y \neq e_1$,

$$y^\top Ay < e_1^\top Ay.$$

By Lemma 1.3, this establishes that e_1 is ESS. By the cyclic symmetry of the game, the same calculation applies to e_2 and e_3 . Therefore,

$$\text{ESS} = \{e_1, e_2, e_3\}.$$

3. Singularities of the replicator dynamics

We now identify the singularities (rest points) of the replicator dynamics

$$\dot{x}_i = x_i((Ax)_i - x^\top Ax), \quad i = 1, 2, 3,$$

on the simplex Δ . Recall that x is a singularity if and only if $\dot{x}_i = 0$ for all i .

(a) Nash equilibria are singularities

Let x^* be one of the symmetric Nash equilibria found above. For any i with $(x^*)_i > 0$ we have, by the definition of a symmetric Nash equilibrium, that strategy i yields the same payoff as any other strategy in the support of x^* . In particular,

$$(Ax^*)_i = x^{*\top} Ax^* \quad \text{whenever } (x^*)_i > 0.$$

Hence, for all i ,

$$\dot{x}_i^* = (x^*)_i((Ax^*)_i - x^{*\top} Ax^*) = 0,$$

so x^* is a rest point of the replicator dynamics.

In our game we therefore know that the seven Nash equilibria

$$e_1, e_2, e_3, \left(\frac{2}{3}, \frac{1}{3}, 0\right), \left(0, \frac{2}{3}, \frac{1}{3}\right), \left(\frac{1}{3}, 0, \frac{2}{3}\right), \left(\frac{1}{3}, \frac{1}{3}, \frac{1}{3}\right)$$

are all singularities.

(b) Other intersections of indifference lines with the edges

Besides these points, each indifference line Z_{ij} also intersects the “opposite” edge of the simplex. We now show that those three intersection points are not singularities.

As an example, consider

$$\bar{x} = \left(0, \frac{1}{3}, \frac{2}{3}\right),$$

which lies on the edge $\{e_2, e_3\}$ and on the indifference line $Z_{1,2}$ (since $x_2 = \frac{1}{3}$). Thus strategy 1 and 2 are indifferent at $\bar{x}$, but strategy 1 is not in the support.

We compute

$$A\bar{x} = \begin{pmatrix} 2 & 0 & 1 \\ 1 & 2 & 0 \\ 0 & 1 & 2 \end{pmatrix} \begin{pmatrix} 0 \\ \frac{1}{3} \\ \frac{2}{3} \end{pmatrix} = \begin{pmatrix} \frac{2}{3} \\ \frac{2}{3} \\ \frac{5}{3} \end{pmatrix},$$

and

$$\phi(\bar{x}) = \bar{x}^\top A\bar{x} = 0 \cdot \frac{2}{3} + \frac{1}{3} \cdot \frac{2}{3} + \frac{2}{3} \cdot \frac{5}{3} = \frac{4}{3}.$$

For $i = 2, 3$ we have $(\bar{x})_i > 0$, but

$$(A\bar{x})_2 = \frac{2}{3} \neq \frac{4}{3}, \quad (A\bar{x})_3 = \frac{5}{3} \neq \frac{4}{3},$$

and hence

$$\dot{x}_2(\bar{x}) = \bar{x}_2((A\bar{x})_2 - \phi(\bar{x})) = \frac{1}{3}\left(\frac{2}{3} - \frac{4}{3}\right) = -\frac{2}{9} \neq 0.$$

Therefore $\bar{x}$ is not a singularity of the replicator dynamics.

The two remaining “opposite” intersection points,

$$\left(\frac{2}{3}, 0, \frac{1}{3}\right) \quad \text{and} \quad \left(\frac{1}{3}, \frac{2}{3}, 0\right),$$

are treated in exactly the same way: in each case one finds that for at least one i with $x_i > 0$ we have $(Ax)_i \neq x^\top Ax$, and thus $\dot{x}_i \neq 0$. Consequently these three points are not rest points.

Hence, for this game, the sets of singularities and Nash equilibria coincide: the only singularities of the replicator dynamics are

$$e_1, e_2, e_3, \left(\frac{2}{3}, \frac{1}{3}, 0\right), \left(0, \frac{2}{3}, \frac{1}{3}\right), \left(\frac{1}{3}, 0, \frac{2}{3}\right), \left(\frac{1}{3}, \frac{1}{3}, \frac{1}{3}\right).$$

4. Linearisation and classification of singularities

In this section we classify the singularities of the replicator dynamics according to the eigenvalues of the linearisation at each rest point. We distinguish three cases: the interior equilibrium, the three two-strategy edge equilibria, and the three pure-strategy equilibria.

(a) Interior singularity $E = (\frac{1}{3}, \frac{1}{3}, \frac{1}{3})$

Let

$$E = \left(\frac{1}{3}, \frac{1}{3}, \frac{1}{3}\right), \quad x = E + h, \quad h_1 + h_2 + h_3 = 0,$$

Where E is the interior singularity we find previously, and h is the disturbance. so that we may eliminate $h_3 = -h_1 - h_2$ and work in the two-dimensional tangent space spanned by (h_1, h_2) .

The replicator dynamics is

$$\dot{x}_i = x_i((Ax)_i - x^\top Ax).$$

Step 1: Expand $x^\top Ax$. We compute

$$x^\top Ax = (E + h)^\top A(E + h) = E^\top AE + E^\top Ah + h^\top AE + h^\top Ah.$$

Since $AE = (1, 1, 1)^\top$, we have

$$h^\top AE = h_1 + h_2 + h_3 = 0.$$

Thus,

$$x^\top Ax = E^\top AE + E^\top Ah + O(\|h\|^2).$$

Step 2: Compute $E^\top Ah$.

We first compute Ah . Using the payoff matrix

$$A = \begin{pmatrix} 2 & 0 & 1 \\ 1 & 2 & 0 \\ 0 & 1 & 2 \end{pmatrix},$$

and $h_3 = -h_1 - h_2$, we obtain

$$(Ah)_1 = 2h_1 + h_3 = h_1 - h_2,$$

$$(Ah)_2 = h_1 + 2h_2,$$

$$(Ah)_3 = h_2 + 2h_3 = -2h_1 - h_2.$$

Therefore

$$E^\top Ah = \frac{1}{3}((Ah)_1 + (Ah)_2 + (Ah)_3).$$

But

$$(Ah)_1 + (Ah)_2 + (Ah)_3 = (h_1 - h_2) + (h_1 + 2h_2) + (-2h_1 - h_2) = 0,$$

hence

$$E^\top Ah = 0.$$

Thus

$$x^\top Ax = E^\top AE + O(\|h\|^2), \quad E^\top AE = 1.$$

Step 3: Expand $(Ax)_i$.

Since

$$Ax = A(E + h) = AE + Ah = (1, 1, 1)^\top + Ah,$$

we have

$$(Ax)_i = 1 + (Ah)_i.$$

Step 4: Substitute into the replicator equation.

Hence, for each i ,

$$\begin{aligned} (Ax)_i - x^\top Ax &= (1 + (Ah)_i) - (E^\top AE + O(\|h\|^2)) \\ &= (Ah)_i + O(\|h\|^2). \end{aligned}$$

Hence

$$\dot{x}_i = x_i((Ax)_i - x^\top Ax) = (E_i + h_i)((Ah)_i + O(\|h\|^2)),$$

so, to first order in h ,

$$\dot{h}_i = \left(\frac{1}{3} + h_i\right)(E_i(Ah)_i) = \frac{1}{3}(Ah)_i + O(h^2).$$

Using the expressions for $(Ah)_1$ and $(Ah)_2$ above, we obtain

$$\begin{pmatrix} \dot{h}_1 \\ \dot{h}_2 \end{pmatrix} = \frac{1}{3} \begin{pmatrix} 1 & -1 \\ 1 & 2 \end{pmatrix} \begin{pmatrix} h_1 \\ h_2 \end{pmatrix} =: J_E \begin{pmatrix} h_1 \\ h_2 \end{pmatrix}.$$

The characteristic polynomial of J_E is

$$\lambda^2 - (\text{tr } J_E)\lambda + \det J_E = \lambda^2 - \lambda + \frac{1}{3} = 0,$$

so

$$\lambda_{1,2} = \frac{1}{2} \pm \frac{i\sqrt{3}}{6}.$$

Both eigenvalues have positive real part $1/2 > 0$, hence the interior rest point E is an *unstable spiral* (repelling focus): nearby trajectories spiral away from E .

Keeping only linear terms in h ,

$$\dot{h}_i = E_i(Ah)_i.$$

Since $E_i = \frac{1}{3}$, we obtain

$$\dot{h}_i = \frac{1}{3}(Ah)_i.$$

Substituting the expressions for Ah ,

$$\begin{pmatrix} \dot{h}_1 \\ \dot{h}_2 \end{pmatrix} = \frac{1}{3} \begin{pmatrix} 1 & -1 \\ 1 & 2 \end{pmatrix} \begin{pmatrix} h_1 \\ h_2 \end{pmatrix}.$$

Thus the Jacobian at E is

$$J_E = \frac{1}{3} \begin{pmatrix} 1 & -1 \\ 1 & 2 \end{pmatrix}.$$

Step 5: Eigenvalues.

The characteristic polynomial is

$$\lambda^2 - (\text{tr } J_E)\lambda + \det J_E = \lambda^2 - \lambda + \frac{1}{3} = 0.$$

Hence

$$\lambda_{1,2} = \frac{1}{2} \pm \frac{i\sqrt{3}}{6}.$$

Both eigenvalues have positive real part $1/2 > 0$, so the interior singularity E is an *repelling*.

(b) Edge singularities: two-strategy mixed equilibria

Edge singularity $E_{12} = (\frac{2}{3}, \frac{1}{3}, 0)$

Here we still use the same method as above. Consider the edge equilibrium

$$E_{12} = \left(\frac{2}{3}, \frac{1}{3}, 0\right).$$

Set

$$E = \left(\frac{2}{3}, \frac{1}{3}, 0\right), \quad x = E + h, \quad h = (h_1, h_2, h_3),$$

and impose the simplex constraint

$$x_1 + x_2 + x_3 = 1 \implies h_1 + h_2 + h_3 = 0,$$

so that we may eliminate $h_3 = -h_1 - h_2$ and work in the two-dimensional tangent space spanned by (h_1, h_2) .

As before,

$$x^\top Ax = (E+h)^\top A(E+h) = E^\top AE + E^\top Ah + h^\top AE + h^\top Ah = E^\top AE + E^\top Ah + h^\top AE + O(\|h\|^2).$$

We first compute

$$AE = \begin{pmatrix} 2 & 0 & 1 \\ 1 & 2 & 0 \\ 0 & 1 & 2 \end{pmatrix} \begin{pmatrix} 2/3 \\ 1/3 \\ 0 \end{pmatrix} = \begin{pmatrix} 4/3 \\ 4/3 \\ 1/3 \end{pmatrix},$$

so that

$$E^\top AE = E^\top (AE) = \frac{2}{3} \cdot \frac{4}{3} + \frac{1}{3} \cdot \frac{4}{3} = \frac{4}{3}.$$

Next we compute Ah . Using $h_3 = -h_1 - h_2$,

$$Ah = \begin{pmatrix} 2 & 0 & 1 \\ 1 & 2 & 0 \\ 0 & 1 & 2 \end{pmatrix} \begin{pmatrix} h_1 \\ h_2 \\ h_3 \end{pmatrix} = \begin{pmatrix} 2h_1 + h_3 \\ h_1 + 2h_2 \\ h_2 + 2h_3 \end{pmatrix} = \begin{pmatrix} h_1 - h_2 \\ h_1 + 2h_2 \\ -2h_1 - h_2 \end{pmatrix}.$$

Hence

$$E^\top Ah = \left(\frac{2}{3}, \frac{1}{3}, 0 \right) \begin{pmatrix} h_1 - h_2 \\ h_1 + 2h_2 \\ -2h_1 - h_2 \end{pmatrix} = h_1,$$

and

$$h^\top AE = h^\top \begin{pmatrix} 4/3 \\ 4/3 \\ 1/3 \end{pmatrix} = \frac{4}{3}(h_1 + h_2) + \frac{1}{3}h_3 = \frac{4}{3}(h_1 + h_2) + \frac{1}{3}(-h_1 - h_2) = h_1 + h_2.$$

Therefore

$$x^\top Ax = E^\top AE + E^\top Ah + h^\top AE + O(\|h\|^2) = \frac{4}{3} + 2h_1 + h_2 + O(\|h\|^2).$$

On the other hand,

$$Ax = A(E + h) = AE + Ah,$$

so for the first two components

$$\begin{aligned} (Ax)_1 &= \frac{4}{3} + (h_1 - h_2), \\ (Ax)_2 &= \frac{4}{3} + (h_1 + 2h_2). \end{aligned}$$

Thus

$$\begin{aligned}
(Ax)_1 - x^\top Ax &= \left(\frac{4}{3} + h_1 - h_2\right) - \left(\frac{4}{3} + 2h_1 + h_2\right) + O(\|h\|^2) \\
&= -h_1 - 2h_2 + O(\|h\|^2), \\
(Ax)_2 - x^\top Ax &= \left(\frac{4}{3} + h_1 + 2h_2\right) - \left(\frac{4}{3} + 2h_1 + h_2\right) + O(\|h\|^2) \\
&= -h_1 + h_2 + O(\|h\|^2).
\end{aligned}$$

Now

$$\dot{x}_i = x_i((Ax)_i - x^\top Ax) = (E_i + h_i)((Ax)_i - x^\top Ax),$$

so the linearisation is obtained by replacing x_i with E_i :

$$\dot{h}_i = E_i((Ax)_i - x^\top Ax)_{\text{linear}}, \quad i = 1, 2.$$

Hence

$$\begin{aligned}
\dot{h}_1 &= E_1(-h_1 - 2h_2) = \frac{2}{3}(-h_1 - 2h_2) = -\frac{2}{3}h_1 - \frac{4}{3}h_2, \\
\dot{h}_2 &= E_2(-h_1 + h_2) = \frac{1}{3}(-h_1 + h_2) = -\frac{1}{3}h_1 + \frac{1}{3}h_2.
\end{aligned}$$

Thus, in matrix form,

$$\begin{pmatrix} \dot{h}_1 \\ \dot{h}_2 \end{pmatrix} = J_{12} \begin{pmatrix} h_1 \\ h_2 \end{pmatrix}, \quad J_{12} = \begin{pmatrix} -\frac{2}{3} & -\frac{4}{3} \\ -\frac{1}{3} & \frac{1}{3} \end{pmatrix}.$$

The trace and determinant are

$$\text{tr } J_{12} = -\frac{1}{3}, \quad \det J_{12} = -\frac{2}{3} < 0,$$

so the eigenvalues are real with opposite signs; indeed

$$\lambda_1 = -1, \quad \lambda_2 = \frac{2}{3}.$$

Hence the edge singularity E_{12} is a *saddle point*. By symmetry and the similar calculation process, we find that they are also saddle with the same eigenvalues holds for the other two edge equilibria E_{23} and E_{31} .

Therefore all three two-strategy mixed equilibria are saddle singularities of the replicator dynamics.

(c) Pure-strategy singularities e_1, e_2, e_3

Finally we consider the three pure strategies

$$e_1 = (1, 0, 0), \quad e_2 = (0, 1, 0), \quad e_3 = (0, 0, 1).$$

Again, by symmetry it suffices to analyse e_1 .

Near e_1 we introduce mutant coordinates

$$x_1 = 1 - y_2 - y_3, \quad x_2 = y_2, \quad x_3 = y_3,$$

with y_2, y_3 small. Substituting into the replicator dynamics and keeping only linear terms in y_2, y_3 (as in the ESS computation) we obtain

$$\dot{y}_2 = -y_2 + O(y^2), \quad \dot{y}_3 = -2y_3 + O(y^2).$$

Thus the linearisation in the tangent space at e_1 is

$$\begin{pmatrix} \dot{y}_2 \\ \dot{y}_3 \end{pmatrix} = J_{e_1} \begin{pmatrix} y_2 \\ y_3 \end{pmatrix}, \quad J_{e_1} = \begin{pmatrix} -1 & 0 \\ 0 & -2 \end{pmatrix},$$

with eigenvalues

$$\lambda_1 = -1, \quad \lambda_2 = -2.$$

Both eigenvalues are strictly negative, hence e_1 is an *attracting node* (asymptotically stable).

By cyclic symmetry and the same calculation process, the same argument applies to e_2 and e_3 , and their linearisations have the same eigenvalues $\{-1, -2\}$ in the 2-dimensional tangent space. Consequently all three pure strategies are attracting singularities of the replicator dynamics.

In summary:

- the interior singularity $(\frac{1}{3}, \frac{1}{3}, \frac{1}{3})$ is an repelling;
- the three two-strategy mixed equilibria on the edges are saddle points;
and
- the three pure strategies e_1, e_2, e_3 are attracting nodes.

This classification is consistent with the phase portrait of the replicator dynamics in the simplex shown below:

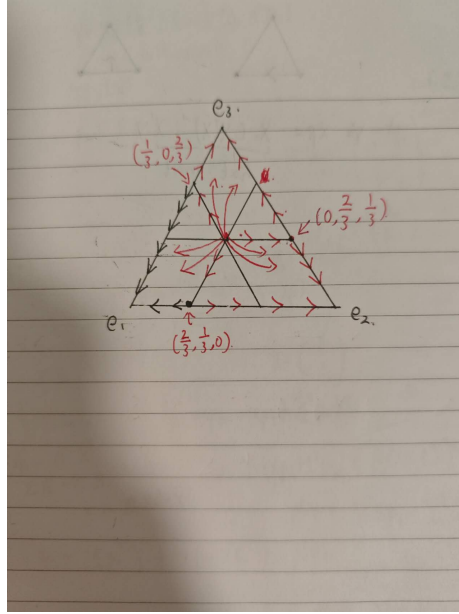


Figure 3: Phase protrait

1 Question 2

We consider the battle of the sexes game

$$A = \begin{pmatrix} 2 & 0 \\ 0 & 2 \end{pmatrix}, \quad B = \begin{pmatrix} 1 & 0 \\ 0 & 7 \end{pmatrix},$$

where A is the payoff matrix of player x and B that of player y . We use the second notation of Section 2.4.

Step 1: Parameters α and β . In the second convention we define

$$\alpha_1 = a_{21} - a_{11}, \quad \alpha_2 = a_{12} - a_{22}, \quad \beta_1 = b_{12} - b_{11}, \quad \beta_2 = b_{21} - b_{22}.$$

For the present game we have

$$a_{11} = 2, \quad a_{12} = 0, \quad a_{21} = 0, \quad a_{22} = 2, \quad b_{11} = 1, \quad b_{12} = 0, \quad b_{21} = 0, \quad b_{22} = 7,$$

hence by using the second convention:

$$\begin{aligned} \alpha_1 &= 0 - 2 = -2, & \alpha_2 &= 0 - 2 = -2, \\ \beta_1 &= 0 - 1 = -1, & \beta_2 &= 0 - 7 = -7. \end{aligned}$$

Step 2: Coordination game. We compute

$$\alpha_1\alpha_2 = (-2)(-2) = 4 > 0, \quad \beta_1\beta_2 = (-1)(-7) = 7 > 0, \quad \alpha_1\beta_1 = (-2)(-1) = 2 > 0.$$

By Proposition 2.1 this implies that the game is a coordination game.

Step 3: Mixed Nash equilibrium via α, β . Let x_2 be the probability that player x plays e_2 and y_2 the probability that player y plays e_2 . In the second convention the interior mixed Nash equilibrium satisfies

$$x_2^* = \frac{\beta_1}{\beta_1 + \beta_2}, \quad y_2^* = \frac{\alpha_1}{\alpha_1 + \alpha_2}.$$

Substituting the values of $\beta_1, \beta_2, \alpha_1, \alpha_2$ we obtain

$$x_2^* = \frac{-1}{-1-7} = \frac{1}{8}, \quad y_2^* = \frac{-2}{-2-2} = \frac{1}{2}.$$

Therefore the mixed Nash equilibrium in terms of the original strategies is

$$x^* = \frac{7}{8}e_1 + \frac{1}{8}e_2, \quad y^* = \frac{1}{2}e_1 + \frac{1}{2}e_2.$$

Step 4: Pure-strategy Nash equilibria via best responses. The payoff bimatrix is

	e_1	e_2
e_1	(2, 1)	(0, 0)
e_2	(0, 0)	(2, 7)

$$BR_A(e_1) = \arg \max_{x \in \Delta} x \cdot A \begin{pmatrix} 1 \\ 0 \end{pmatrix} = \arg \max_{x \in \Delta} (2x_1 + 0x_2) = \{e_1\}.$$

$$BR_A(e_2) = \arg \max_{x \in \Delta} x \cdot A \begin{pmatrix} 0 \\ 1 \end{pmatrix} = \arg \max_{x \in \Delta} (0x_1 + 2x_2) = \{e_2\}.$$

$$BR_B(e_1) = \arg \max_{y \in \Delta} By \cdot \begin{pmatrix} 1 \\ 0 \end{pmatrix} = \arg \max_{y \in \Delta} (1y_1 + 0y_2) = \{e_1\}.$$

$$BR_B(e_2) = \arg \max_{y \in \Delta} By \cdot \begin{pmatrix} 0 \\ 1 \end{pmatrix} = \arg \max_{y \in \Delta} (0y_1 + 7y_2) = \{e_2\}.$$

Therefore, the game has two pure Nash equilibria (e_1, e_1) and (e_2, e_2) , and one mixed Nash equilibrium

$$\left(\frac{7}{8}e_1 + \frac{1}{8}e_2, \frac{1}{2}e_1 + \frac{1}{2}e_2\right).$$

Property of those three points and proof We write the replicator dynamics in the second notation as

$$\dot{x}_2 = x_2(1 - x_2)[-2 + 4y_2], \quad \dot{y}_2 = y_2(1 - y_2)[-1 + 8x_2],$$

where we used $\alpha_1 = \alpha_2 = -2$ and $\beta_1 = -1, \beta_2 = -7$. Let $x := x_2$ and $y := y_2$, and define

$$f(x, y) = x(1 - x)(-2 + 4y), \quad g(x, y) = y(1 - y)(-1 + 8x).$$

The Jacobian matrix of the vector field (f, g) is

$$J(x, y) = \begin{pmatrix} \partial_x f(x, y) & \partial_y f(x, y) \\ \partial_x g(x, y) & \partial_y g(x, y) \end{pmatrix}.$$

A short computation gives

$$\begin{aligned} \partial_x f(x, y) &= (1 - 2x)(-2 + 4y), & \partial_y f(x, y) &= 4x(1 - x), \\ \partial_x g(x, y) &= 8y(1 - y), & \partial_y g(x, y) &= (1 - 2y)(-1 + 8x). \end{aligned}$$

The pure equilibria. The two pure Nash equilibria correspond to the corners $(e_1, e_1) = (0, 0)$ and $(e_2, e_2) = (1, 1)$ in (x, y) -coordinates. At $(0, 0)$ we obtain

$$J(0, 0) = \begin{pmatrix} -2 & 0 \\ 0 & -1 \end{pmatrix},$$

whose eigenvalues are -2 and -1 . Both are negative, hence (e_1, e_1) is a locally asymptotically stable equilibrium (an attractive node). At $(1, 1)$ we get

$$J(1, 1) = \begin{pmatrix} -2 & 0 \\ 0 & -7 \end{pmatrix},$$

with eigenvalues -2 and -7 , again both negative. Thus (e_2, e_2) is also locally asymptotically stable.

The mixed equilibrium. The interior mixed Nash equilibrium occurs at

$$x^* = \frac{1}{8}, \quad y^* = \frac{1}{2}.$$

Evaluating the Jacobian at this point yields

$$J\left(\frac{1}{8}, \frac{1}{2}\right) = \begin{pmatrix} 0 & \frac{7}{16} \\ 2 & 0 \end{pmatrix}.$$

Its trace is zero and its determinant is $\det J = -\frac{7}{8} < 0$, so the eigenvalues are

$$\lambda_{\pm} = \pm \sqrt{\frac{7}{8}},$$

one positive and one negative. Therefore the interior mixed equilibrium is a saddle point: it is unstable, with a one-dimensional stable manifold and a one-dimensional unstable manifold separating the basins of attraction of the two pure equilibria (e_1, e_1) and (e_2, e_2) .

Stability of the equilibria. The two pure Nash equilibria (e_1, e_1) and (e_2, e_2) are asymptotically stable under the replicator dynamics. Indeed, near (e_1, e_1) we have $\dot{x}_2 < 0$ and $\dot{y}_2 < 0$, so all trajectories in a neighbourhood are pushed toward the corner; similarly, near (e_2, e_2) we have $\dot{x}_2 > 0$ and $\dot{y}_2 > 0$, so trajectories converge to (e_2, e_2) . Thus both pure equilibria are attractive.

The interior mixed Nash equilibrium is located at

$$x_2^* = \frac{1}{8}, \quad y_2^* = \frac{1}{2}.$$

Linearizing the replicator vector field at this point yields a Jacobian matrix with one positive and one negative eigenvalue. Consequently, the mixed equilibrium is a saddle point: trajectories on its one-dimensional stable manifold converge to it, whereas all other nearby trajectories are repelled away toward the two attractive pure equilibria. Thus the orbit is shown below.

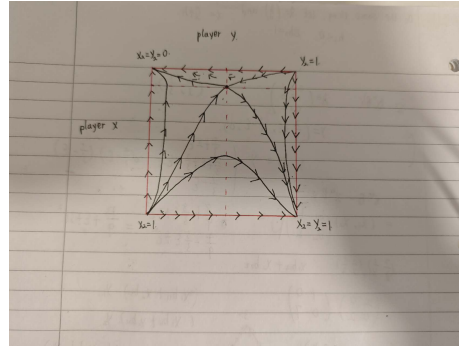


Figure 4: Orbit

2 Question 3

(a) **There exists no NE in the interior of $\Delta_A \times \Delta_B$**

Let $q = (q_1, q_2, q_3) \in \Delta_B$ be player B's mixed strategy. Player A's expected payoffs from rows 1, 2, 3 are given by the components of Aq , and since

$$A = \begin{pmatrix} 1 & 0 & -1 \\ -1 & 0 & -1 \\ 1 & 0 & -2 \end{pmatrix}, \quad Aq = \begin{pmatrix} q_1 - q_3 \\ -q_1 - q_3 \\ q_1 - 2q_3 \end{pmatrix}.$$

An interior Nash equilibrium requires that A be indifferent between all three rows, so

$$(Aq)_1 = (Aq)_2 = (Aq)_3.$$

Indifference between rows 1 and 2.

$$q_1 - q_3 = -q_1 - q_3 \quad \Rightarrow \quad 2q_1 = 0 \quad \Rightarrow \quad q_1 = 0.$$

Indifference between rows 1 and 3.

$$q_1 - q_3 = q_1 - 2q_3 \Rightarrow q_3 = 0.$$

Because $q_1 + q_2 + q_3 = 1$, we obtain

$$q = (0, 1, 0),$$

which lies on the boundary of Δ_B . Thus no interior q can make A indifferent among all three rows.

Now let $p = (p_1, p_2, p_3) \in \Delta_A$ be player A's mixed strategy. B's expected payoffs from columns are given by the vector pB . Since

$$B = \begin{pmatrix} 1 & -1 & 1 \\ 0 & 0 & 0 \\ -1 & -1 & -2 \end{pmatrix}, \quad pB = \begin{pmatrix} p_1 - p_3 \\ -p_1 - p_3 \\ p_1 - 2p_3 \end{pmatrix}.$$

B's indifference among all three columns requires

$$(pB)_1 = (pB)_2 = (pB)_3.$$

Following the same calculations as for A, we obtain

$$p_1 = 0, \quad p_3 = 0,$$

and hence

$$p = (0, 1, 0),$$

which lies on the boundary of Δ_A .

Since neither player can be indifferent among all three strategies at an interior point of their simplex, there is **no Nash equilibrium in the interior of $\Delta_A \times \Delta_B$** .

(b) The NE set contains the line segment $\{e_1\} \times [e_1, e_3]$

Fix first player A's strategy to be $p = e_1 = (1, 0, 0)$. Player B's expected payoff from a mixed strategy $q \in \Delta_B$ is

$$u_B(q) = pBq = e_1Bq = (1, -1, 1) \cdot q = q_1 - q_2 + q_3.$$

Since $q_1 + q_2 + q_3 = 1$, this can be written as

$$u_B(q) = 1 - 2q_2,$$

which is strictly decreasing in q_2 and not relate to the value of q_1 or q_3 . Thus B maximises $u_B(q)$ by choosing $q_2 = 0$. Hence

$$\text{BR}_B(e_1) = \{q \in \Delta_B : q_2 = 0\} = [e_1, e_3] = \{\alpha e_1 + (1 - \alpha)e_3 : \alpha \in [0, 1)\}.$$

Now consider A's best replies against any $q \in [e_1, e_3]$. Write such a q as

$$q = \alpha e_1 + (1 - \alpha)e_3, \quad \alpha \in [0, 1].$$

Then A's row payoffs are the components of Aq :

$$Aq = \begin{pmatrix} \alpha - (1 - \alpha) \\ -\alpha - (1 - \alpha) \\ \alpha - 2(1 - \alpha) \end{pmatrix} = \begin{pmatrix} 2\alpha - 1 \\ -1 \\ 3\alpha - 2 \end{pmatrix}.$$

For $0 \leq \alpha \leq 1$, we have

$$e_1 Aq = 2\alpha - 1 \geq -1 = e_2 Aq, \quad e_1 Aq = 2\alpha - 1 \geq 3\alpha - 2 = e_3 Aq,$$

so row 1 strictly maximises A's payoff. Thus,

$$BR_A(q) = \{e_1\} \quad \text{for all } q \in [e_1, e_3].$$

Combining this with $BR_B(e_1) = [e_1, e_3]$, every pair

$$(e_1, q), \quad q \in [e_1, e_3],$$

is a Nash equilibrium. Thus the NE set of the game contains the line segment

$$\{e_1\} \times [e_1, e_3].$$

(c) The Nash equilibrium set as a union of six line segments

From part (a) we know that no interior Nash equilibrium exists. Thus every equilibrium (p, q) must lie on the boundary of Δ_A and/or Δ_B . We now determine all such boundary equilibria.

Step 1. Best responses of A. For $q = (q_1, q_2, q_3) \in \Delta_B$ we computed

$$Aq = \begin{pmatrix} q_1 - q_3 \\ -q_1 - q_3 \\ q_1 - 2q_3 \end{pmatrix}.$$

(i) q in the open edge $[e_1, e_3]$ (so $q_2 = 0$ and $q_1, q_3 > 0$). Since $q_1 + q_3 = 1$,

$$u_A(1, q) = q_1 - q_3 = 1 - 2q_3, \quad u_A(2, q) = -1, \quad u_A(3, q) = q_1 - 2q_3 = (1 - q_3) - 2q_3 = 1 - 3q_3.$$

Hence $u_A(1, q) - u_A(3, q) = q_3 > 0$ and $u_A(2, q) = -1 < 1 - 2q_3$. Therefore

$$BR_A(q) = \{e_1\}.$$

(ii) q in the open edge $[e_1, e_2]$ (so $q_3 = 0$ and $q_1 > 0$). Then

$$Aq = (q_1, -q_1, q_1),$$

and again $u_A(1, q) = u_A(3, q) > u_A(2, q)$, so

$$BR_A(q) = \{[e_1, e_3]\}.$$

(iii) q in the edge $[e_2, e_3]$ (so $q_1 = 0$ and $q_3 \geq 0$). Then

$$Aq = (-q_3, -q_3, -2q_3),$$

and $u_A(1, q) = u_A(2, q) > u_A(3, q)$, so

$$BR_A(q) = [e_1, e_2].$$

(iv) **Vertices.**

$$Ae_1 = (1, -1, 1), \quad Ae_2 = (0, 0, 0), \quad Ae_3 = (-1, -1, -2),$$

so

$$BR_A(e_1) = [e_1, e_3], \quad BR_A(e_2) = \Delta_A, \quad BR_A(e_3) = [e_1, e_2].$$

Step 2. Best responses of B. For $p = (p_1, p_2, p_3) \in \Delta_A$ we have

$$pB = \begin{pmatrix} p_1 - p_3 \\ -p_1 - p_3 \\ p_1 - 2p_3 \end{pmatrix},$$

which is algebraically identical to Aq with (q_1, q_3) replaced by (p_1, p_3) . Thus the above analysis applies symmetrically:

$$\begin{aligned} p \in [e_1, e_3] &\implies BR_B(p) = \{e_1\}, \\ p \in [e_1, e_2] &\implies BR_B(p) = \{e_1\}, \\ p \in [e_2, e_3] &\implies BR_B(p) = [e_1, e_2], \\ p = e_1 &\implies BR_B(e_1) = [e_1, e_3], \\ p = e_2 &\implies BR_B(e_2) = \Delta_B, \\ p = e_3 &\implies BR_B(e_3) = [e_1, e_2]. \end{aligned}$$

Step 3. Solving the fixed-point conditions. A pair (p, q) is a Nash equilibrium iff

$$p \in BR_A(q) \quad \text{and} \quad q \in BR_B(p).$$

We now examine each boundary region using the correct best responses above. By using the same method used in part (b), we can easily conclude the following six segment are NE segment:

$$S_1 = \{e_1\} \times [e_1, e_3].$$

$$S_2 = [e_1, e_3] \times \{e_1\}.$$

Here S_1 and S_2 is the line segment we prove in part (b), and because of symmetry. We can directly get S_2 .

Then, we have: The segment

$$\{e_3\} \times [e_1, e_2] = \{(p, q) \in \Delta_A \times \Delta_B : p = e_3, q = \mu e_1 + (1 - \mu)e_2, \mu \in [0, 1]\}$$

consists entirely of Nash equilibria.

Take any

$$q = \mu e_1 + (1 - \mu)e_2, \quad \mu \in [0, 1].$$

Then

$$Aq = A(\mu e_1 + (1 - \mu)e_2) = \mu Ae_1 + (1 - \mu)Ae_2 = \begin{pmatrix} \mu \\ -\mu \\ \mu \end{pmatrix}.$$

For $\mu > 0$, the first and third components are equal and strictly greater than the second:

$$(Aq)_1 = (Aq)_3 = \mu > -\mu = (Aq)_2.$$

For $\mu = 0$, all three components coincide. Hence for all $\mu \in [0, 1]$,

$$BR_A(q) = [e_1, e_3] \supset \{e_3\}.$$

Next consider player B's best response to $p = e_3$. Since

$$p^\top B = e_3^\top B = (-1, -1, -2),$$

we obtain

$$BR_B(e_3) = [e_1, e_2].$$

Therefore, for every $\mu \in [0, 1]$,

$$e_3 \in BR_A(\mu e_1 + (1 - \mu)e_2), \quad \mu e_1 + (1 - \mu)e_2 \in BR_B(e_3),$$

which shows that $(e_3, \mu e_1 + (1 - \mu)e_2)$ is a Nash equilibrium. Thus the entire segment $\{e_3\} \times [e_1, e_2]$ consists of Nash equilibria. And due to the symmetry of the payoff matrix, we can conclude $[e_1, e_2] \times \{e_3\}$ is also a line segment. Hence we get the following:

$$S_3 = \{e_3\} \times [e_1, e_2].$$

$$S_4 = \{e_3\} \times [e_1, e_2].$$

Finally, by using the same method above, we can find The line segment

$$\{e_2\} \times [e_2, e_3] = \{(p, q) \in \Delta_A \times \Delta_B : p = e_2, q = \mu e_2 + (1 - \mu)e_3, \mu \in [0, 1]\}$$

consists entirely of Nash equilibria. Fix any

$$q = \mu e_2 + (1 - \mu)e_3, \quad \mu \in [0, 1].$$

Then

$$Aq = \mu Ae_2 + (1 - \mu)Ae_3 = \mu \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix} + (1 - \mu) \begin{pmatrix} -1 \\ -1 \\ -2 \end{pmatrix} = \begin{pmatrix} -1 + \mu \\ -1 + \mu \\ -2 + 2\mu \end{pmatrix}.$$

For $\mu \in [0, 1)$ we have

$$-1 + \mu > -2 + 2\mu,$$

so the first and second components of Aq are equal and strictly larger than the third; hence

$$BR_A(q) = [e_1, e_2] \supset \{e_2\}.$$

For $\mu = 1$ we get $q = e_2$ and $Aq = Ae_2 = (0, 0, 0)^\top$, so all three pure strategies of player A are best responses, and again $e_2 \in BR_A(q)$. Thus, for every $q \in [e_2, e_3]$ we have

$$e_2 \in BR_A(q).$$

Next, consider player B's best response to $p = e_2$. Since

$$e_2^\top B = (0, 0, 0),$$

all of B's pure strategies yield the same payoff against e_2 , and hence

$$BR_B(e_2) = \Delta_B \supset [e_2, e_3].$$

Therefore, for every $\mu \in [0, 1]$,

$$e_2 \in BR_A(\mu e_2 + (1 - \mu)e_3), \quad \mu e_2 + (1 - \mu)e_3 \in BR_B(e_2),$$

which shows that each pair $(e_2, \mu e_2 + (1 - \mu)e_3)$ is a Nash equilibrium. Hence the entire line segment $\{e_2\} \times [e_2, e_3]$ consists of Nash equilibria.

$$S_5 = [e_2, e_3] \times \{e_2\}, \quad S_6 = \{e_2\} \times [e_2, e_3]$$

Conclusion. Collecting all valid pairs, the full Nash equilibrium set is exactly

$$NE(A, B) = S_1 \cup S_2 \cup S_3 \cup S_4 \cup S_5 \cup S_6,$$

six line segments forming a closed loop in the boundary of $\Delta_A \times \Delta_B$.